**9** Fig. 9.1 shows a circuit with a 3-position switch.

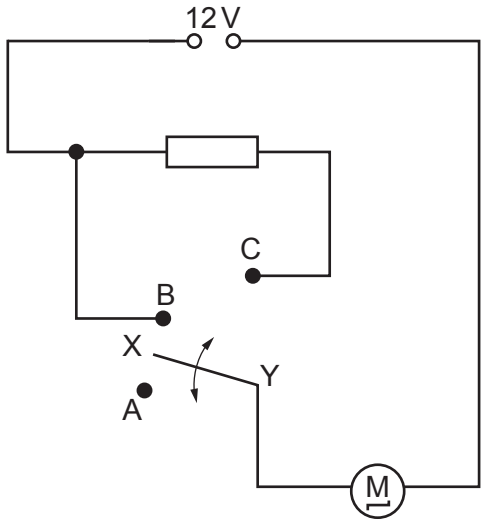



**Fig. 9.1**

The moving part of the switch is always connected to point Y around which it pivots. The other end of the moving part, labelled X, can be connected to one of the points A, B or C.

**(a)** The resistance of the motor is 2.0 Ω and the resistance of the resistor is 3.0 Ω.

Determine the current in the motor when the switch is connected to:

**(i)** point A

current = ..... [1]

**(ii)** point B

current = ..... [2]

**(iii)** point C.

current = ..... [2]

**(b)** Two resistors of resistance 2.0 Ω and 3.0 Ω are connected in parallel.

Calculate the combined resistance of the resistors in this arrangement.

resistance = ..... [3]

[Total: 8]

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